HW #6

Thursday, March 26, 2026 4:06 PM

P8-5. All calculations solved in Python

a) 6.25 hr in US

b) 7.81 hr in Sweden

c) 10.4 hr in Russia

d) 11.5 hr in US, 13.0 hr in Sweden, 15.6 hr in Russia ?

e) 6.25 hr in US, 7.81 hr in Sweden, 10.4 hr in Russia

f) In the US, you have 187.2 min to reach home

In Sweden, 105.5 min

In Russia, 0 min

g) The alcohol concentration would rise more gradually & until 2 hr

h) Alcohol in blood content would increase much faster for a thin person but a heavy person would retain a high BAC for longer.

i) People can drink the same amount of alcohol but the time spent drinking will affect drunkness.

P8-7. a) Semibatch, high temp

b) PFR, low temp

c) PFR, keep C_B low

d) PFR, low temp

e) PFR, high temp

f) Semibatch, high temp

P8-11. a) $r_{1A} = -k_1 C_A$

$= -(7\ 1/\text{min})(0.1\ \text{mol/dm}^3)$

$r_{1A} = -0.7\ \text{mol/dm}^3\cdot\text{min}$

$r_{1A} = -0.7 \text{ mol/dm}^3\cdot\text{min}$

$r_{2A} = -\frac{1}{3} r_{2D} = -\frac{1}{3} k_2 C_C^{\,2} C_A$

$= -\frac{1}{3}(3 \text{ dm}^6/\text{mol}^2\cdot\text{min})(0.51 \text{ mol/dm}^3)^2(0.1 \text{ mol/dm}^3)$

$r_{2A} = -0.0260 \text{ mol/dm}^3\cdot\text{min}$

$r_{3A} = 0$

b) $r_{1B} = -\frac{1}{3} r_{1A}$

$r_{1B} = 0.233 \text{ mol/dm}^3\cdot\text{min}$

$r_{2B} = 0$

$r_{3B} = 0$

c) $r_{1C} = -\frac{1}{3} r_{1A}$

$r_{1C} = 0.233 \text{ mol/dm}^3\cdot\text{min}$

$r_{2C} = -\frac{2}{3} r_{2D} = 2 r_{2A}$

$= 2(-0.0260)$

$r_{2C} = -0.0520 \text{ mol/dm}^3\cdot\text{min}$

$r_{3C} = -r_{3E} = -k_{3E} C_D C_C$

$= -(2 \text{ dm}^3/\text{mol}\cdot\text{min})(0.049 \text{ mol/dm}^3)(0.51 \text{ mol/dm}^3)$

$r_{3C} = -0.050 \text{ mol/dm}^3\cdot\text{min}$

d) $r_{1D} = 0$

$r_{2D} = k_{2D} C_C^{\,2} C_A$

$= (3 \text{ dm}^6/\text{mol}^2\cdot\text{min})(0.51 \text{ mol/dm}^3)^2(0.1 \text{ mol/dm}^3)$

$r_{2D} = 0.078 \text{ mol/dm}^3\cdot\text{min}$

$r_{3D} = -\frac{4}{3} r_E = -\frac{4}{3} k_{3E} C_D C_C$

$= -\frac{4}{3}(2 \text{ dm}^3/\text{mol}\cdot\text{min})(0.049 \text{ mol/dm}^3)(0.51 \text{ mol/dm}^3)$

$r_{3D} = -0.067 \text{ mol/dm}^3\cdot\text{min}$

e) $r_{1E} = 0$

$r_{2E} = 0$

$r_{3E} = k_{3E} C_D C_C = (2 \text{ dm}^3/\text{mol}\cdot\text{min})(0.049 \text{ mol/dm}^3)(0.51 \text{ mol/dm}^3)$

$r_{3E} = 0.050 \text{ mol/dm}^3\cdot\text{min}$

f) $r_A = -0.7 - 0.026 \rightarrow r_A = -0.726 \text{ mol/dm}^3\cdot\text{min}$

$r_B = 0.233 \text{ mol/dm}^3\cdot\text{min}$

f) $r_A = -0.1 - 0.026 \rightarrow r_A = -0.126 \text{ mol/dm}^3\cdot\text{min}$

$r_B = 0.233 \text{ mol/dm}^3\cdot\text{min}$

$r_C = 0.233 - 0.052 - 0.050 \rightarrow r_C = 0.131 \text{ mol/dm}^3\cdot\text{min}$

$r_D = 0.078 - 0.067 \rightarrow r_D = 0.011 \text{ mol/dm}^3\cdot\text{min}$

$r_E = 0.050 \text{ mol/dm}^3\cdot\text{min}$

g) $V = \dfrac{F_{A0} - F_{A0} C_A}{-r_A} = \dfrac{(100)(3) - (100)(0.1)}{0.726}$

$V = 400 \text{ dm}^3$

h) $F_A = (100)(0.1) \rightarrow F_A = 10 \text{ mol/min}$

$F_B = (100)(0.93) \rightarrow F_B = 93 \text{ mol/min}$

$F_C = (100)(0.51) \rightarrow F_C = 51 \text{ mol/min}$

$F_D = (100)(0.049) \rightarrow F_D = 4.9 \text{ mol/min}$

$F_E = (400)(0.05) \rightarrow F_E = 20 \text{ mol/min}$

i) $\tilde{S}_{D/E} = 0.52$

$\tilde{S}_{C/D} = 1050$

j) $\tilde{S}_{D/E} = 56.2$

$\tilde{S}_{C/D} = 1.9 \times 10^{-10}$

$\tilde{S}_{D/E}$ is larger than $\tilde{S}_{C/D}$ in part i) but smaller in part j) due to the diffusion of C.